

1- Prove that any bipartite graph with  $n$  vertices has at most  $\frac{n^2}{4}$  edges. (9 points)

A  $|A| = a$        $a + b = n$       at most  $a \cdot b$  edges

B  $|B| = b$        $b = n - a$        $ab = a(n - a)$  sub  $b = n - a$

$$a(n - a)$$

$$f(a) = a(n - a)$$

$$f'(a) = n - 2a$$

$$n - 2a = 0 \quad 2a = n$$

$$a = \frac{n}{2} \Leftrightarrow b = \frac{n}{2} \Rightarrow \left(\frac{n}{2}\right)\left(\frac{n}{2}\right) = \frac{n^2}{4}$$

$\therefore$  Any bipartite graph on  $n$  vertices has at most  $\frac{n^2}{4}$

2- Prove that any bipartite planar graph with  $n$  vertices has at most  $2n - 4$  edges. (12 points)

Euler's Formula:

$$n - e + f = 2, \text{ where } \begin{array}{l} n = \text{vertices} \\ e = \text{edges} \\ f = \text{faces} \end{array}$$

Bipartite graph: no odd cycle

$\hookrightarrow$  cycle =  $n \geq 4$

Each face is bounded by at least 4 edges

Note: each edge is counted twice  
 $\sum \text{face edges} = 2e$

$$2e \geq 4f \quad = \quad e \geq 2f \quad \text{or} \quad \frac{1}{2}e \geq f$$

$$\Rightarrow n - e + \frac{1}{2}e \geq 2 \quad \text{sub. } \frac{1}{2}e \geq f$$

$$n - \frac{1}{2}e \geq 2$$

$$n - 2 \geq \frac{1}{2}e \quad \text{sub. } -2, +\frac{1}{2}e$$

$$2n - 4 \geq e \quad \text{is}$$

(3 faces)

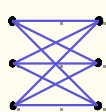
$$\text{Normal planar} = e \leq 3n - 6$$

$\therefore$  Any bipartite planar graph on  $n$  vertices has at most  $2n - 4$  edges

3- Prove that the complete bipartite graph  $K_{3,3}$  is not planar. (6 points)

Lecture 16

$K_{3,3}$



$n = 6$  vertices, from Q2:  $e \leq 2n - 4$

$$e = 9 \text{ edges}$$

$$9 \leq 2(6) - 4$$

$$9 \leq 12 - 4$$

$$9 \leq 8$$

$\therefore$  Contradiction, also a known theorem called Kuratowski's Theorem  $K_5$  or  $K_{3,3}$

4 - A graph is "triangle-free" if it does not contain any cycle of length three (a cycle with three edges).

Prove that any triangle-free planar graph can be colored by 4 colors. (14 points)

Base Case:  $f$  is trivial

Let  $G$  be a triangle-free planar graph with  $n \geq 4$

$$2e \geq 4f \Rightarrow \frac{1}{2}e \geq f \text{ or } e \geq 2f$$

$$n - e + f = 2$$

$$n - e + \frac{1}{2}e \geq 2$$

$$n - \frac{1}{2}e \geq 2 = \frac{e \leq 2n - 4}{\uparrow}$$

$G$  is satisfied

Assume for contradiction every vertex has degree atleast 4

$$2e = \sum_{v \in V} \deg(v) \geq 4n \Rightarrow e \geq 2n$$

$$\underline{2n \leq e \leq 2n - 4}$$

Contradiction for  $n \geq 2$

$G$  must have a vertex  $v$   $\deg(v) \leq 3$

If vertex  $v$  has at most 3 neighbors in  $G$ , this means it needs

at most 3 distinct colors, and if we have colour  $\{1, 2, 3, 4\}$  there exists at least one colour from  $\{1, 2, 3, 4\}$  not used by any neighbor of  $v$ .

Also using the Grotzsch Theorem, we know that every triangle-free planar graph is 3-colorable  
Since its 3 colourable, its also 4 colourable ( $3 \leq 4$ )

5 - Prove that in any group of two or more people, there are two that have the same number of friends in the group. (represent each person by a vertex of a graph and their friendship by an edge) (9 points)

vertex = a person

edge = Connection

domain:  $0, 1, 2, \dots, n-1$ , where  $n-1$  knows everyone, so we can never have 0, and  $n-1$ , because  $n-1$  does not in fact know everyone

Pigeonhole theorem:  $n$  vertices (people)

only  $n-1$  possible degree values

by pigeonhole theorem atleast two vertices must have the same degree

$\therefore$  In any group of two or more people, there are two with the same number of friends